

Disease Mechanisms — GT Precedent Matching

40 diseases | nano-rare-gt | 2026-06-21

Row colour: compatible conditional uncertain

Disease	ORPHA ID	Gene	Mechanism	GT Compat.	Preferred Modality	Mechanism Detail	Evidence Status
Hemophilia B	ORPHA:306	F9	loss_of_function	compatible	gene_addition_or_factor_expres sion	Factor IX deficiency from pathogenic F9 variants	source_linked_needs_ review
Leber congenital amaurosis	ORPHA:65	RPE65	loss_of_function	compatible	gene_addition	RPE65 retinoid-cycle enzyme deficiency	source_linked_needs_ review
Severe combined immunodeficiency due to adenosine deaminase deficiency	ORPHA:277	ADA	loss_of_function	compatible	ex_vivo_hsc_gene_addition	ADA enzyme deficiency causing toxic purine metabolite accumulation	source_linked_needs_ review
Spinal Muscular Atrophy	ORPHA:70	SMN1	loss_of_function	compatible	gene_addition	SMN protein deficiency from biallelic SMN1 mutation	source_linked_needs_ review
Duchenne muscular dystrophy	ORPHA:98896	DMD	loss_of_function_oversiz ed	conditional	engineered_microgene_or_dual_ vector	Dystrophin loss with native DMD too large for single AAV	source_linked_needs_ review
Leber hereditary optic neuropathy	ORPHA:104	MT-ND4	mitochondrial_loss_of_fu nction	uncertain	specialized_allotopic_expres sion_or_mitochondrial_strategy	Mitochondrial complex I dysfunction from MT-ND4 variant	needs_manual_review
Achromatopsia	ORPHA:49382	CNGB3	loss_of_function	compatible	retinal_gene_addition	Cone cyclic nucleotide-gated channel beta-subunit loss	source_linked_needs_ review
Alpha-mannosidosis	ORPHA:61	MAN2B1	loss_of_function	compatible	gene_addition_or_cross_correct ion	Lysosomal alpha-mannosidase enzyme deficiency	source_linked_needs_ review
Choroideremia	ORPHA:180	CHM	loss_of_function	compatible	retinal_gene_addition	REP1 deficiency affecting retinal cells	source_linked_needs_ review
Crigler-Najjar syndrome type I	ORPHA:1060	UGT1A1	loss_of_function	compatible	liver_gene_addition	Absent or severe deficiency of bilirubin UDP-glucuronosyltransferase	source_linked_needs_ review
Fabry disease	ORPHA:324	GLA	loss_of_function	compatible	gene_addition_or_cross_correct ion	Alpha-galactosidase A lysosomal enzyme deficiency	source_linked_needs_ review
Gaucher disease	ORPHA:355	GBA	loss_of_function	conditional	gene_addition_or_cross_correct ion	Beta-glucocerebrosidase lysosomal enzyme deficiency	source_linked_needs_ review
Glycogen storage disease type Ia	ORPHA:79258	G6PC	loss_of_function	compatible	liver_kidney_gene_addition	Glucose-6-phosphatase catalytic subunit deficiency	source_linked_needs_ review
Kohlschutter-Tonz syndrome	ORPHA:1946	ROGDI	loss_of_function	conditional	cns_gene_addition_pending_m echanism_validation	ROGDI loss of function — gene addition is plausible by genetics but molecular function is not fully resolved and disease biology has not been established	source_checked
Krabbe disease	ORPHA:487	GALC	loss_of_function	conditional	gene_addition_or_hsc_cross_c orrection	Galactocerebrosidase lysosomal enzyme deficiency	source_linked_needs_ review
Maple syrup urine disease	ORPHA:511	BCKDHA	loss_of_function	conditional	liver_gene_addition_or_enzyme_ complex_rescue	Branched-chain alpha-ketoacid dehydrogenase E1 alpha deficiency	source_linked_needs_ review
Metachromatic leukodystrophy	ORPHA:512	ARSA	loss_of_function	compatible	ex_vivo_hsc_or_cns_gene_addi tion	Arylsulfatase A lysosomal enzyme deficiency	source_linked_needs_ review
Mucopolidosis type IV	ORPHA:578	MCOLN1	loss_of_function	conditional	direct_cns_aav_per_cell_deliver y	TRPML1 lysosomal membrane Ca2+ channel deficiency — cell-autonomous membrane protein NOT a secretable enzyme	source_linked_needs_ review
Mucopolysaccharidosis type I	ORPHA:579	IDUA	loss_of_function	compatible	gene_addition_or_cross_correct ion	Alpha-L-iduronidase lysosomal enzyme deficiency	source_linked_needs_ review
Mucopolysaccharidosis type II	ORPHA:580	IDS	loss_of_function	compatible	gene_addition_or_cross_correct ion	Iduronate-2-sulfatase lysosomal enzyme deficiency	source_linked_needs_ review
Mucopolysaccharidosis type IIIA (Sanfilippo A)	ORPHA:79269	SGSH	loss_of_function	compatible	cns_gene_addition_or_cross_c orrection	Heparan N-sulfatase lysosomal enzyme deficiency	source_linked_needs_ review
Ornithine transcarbamylase deficiency	ORPHA:664	OTC	loss_of_function	compatible	liver_gene_addition	Ornithine transcarbamylase urea-cycle enzyme deficiency	source_linked_needs_ review
Phenylketonuria	ORPHA:716	PAH	loss_of_function	compatible	liver_gene_addition	Phenylalanine hydroxylase enzyme deficiency	source_linked_needs_ review

Disease	ORPHA ID	Gene	Mechanism	GT Compat.	Preferred Modality	Mechanism Detail	Evidence Status
Pompe disease	ORPHA:365	GAA	loss_of_function	conditional	gene_addition_or_secreted_enzyme_rescue	Acid alpha-glucosidase lysosomal enzyme deficiency	source_linked_needs_review
Salla disease	ORPHA:309	SLC17A5	loss_of_function	conditional	direct_cns_aav_per_cell_delivery	Sialin lysosomal membrane transporter deficiency — cell-autonomous membrane protein NOT a secretable enzyme	source_linked_needs_review
Vitamin B12-unresponsive methylmalonic acidemia	ORPHA:27	MUT	loss_of_function	conditional	liver_aav_with_mts_validation	Methylmalonyl-CoA mutase — nuclear-encoded mitochondrial MATRIX enzyme requiring post-translational import via intact N-terminal MTS	source_linked_needs_review
Wiskott-Aldrich syndrome	ORPHA:906	WAS	loss_of_function	compatible	ex_vivo_hsc_gene_addition	WASP deficiency in hematopoietic cells	source_linked_needs_review
X-linked adrenoleukodystrophy	ORPHA:43	ABCD1	loss_of_function	conditional	ex_vivo_hsc_gene_addition	Peroxisomal ABCD1 transporter deficiency	source_linked_needs_review
X-linked myotubular myopathy	ORPHA:596	MTM1	loss_of_function	compatible	muscle_gene_addition	Myotubularin deficiency in skeletal muscle	source_linked_needs_review
X-linked retinoschisis	ORPHA:792	RS1	loss_of_function	compatible	retinal_gene_addition	Retinoschisin deficiency affecting retinal architecture	source_linked_needs_review
CHARGE syndrome	ORPHA:138	CHD7	haploinsufficiency	conditional	gene_addition_haploinsufficiency_rescue	Loss of one functional CHD7 allele reduces chromatin remodeller dosage below the threshold needed for normal embryonic development; gene addition in principle restores dosage but CHD7 is large (7950bp) and developmental timing and cell-level expression requirements are complex	source_checked
Neurofibromatosis type 1	ORPHA:636	NF1	haploinsufficiency	conditional	gene_addition_tumour_suppressor_rescue	Germline NF1 mutation reduces neurofibromin tumour suppressor dosage; subsequent somatic loss of the wild-type allele in individual cells initiates benign neurofibromas; gene addition cannot prevent independent somatic second-hit events across the lifetime of the patient	source_checked
Friedreich ataxia	ORPHA:95	FXN	repeat_expansion_silencing	conditional	frataxin_gene_addition_or_epigenetic_derepression	GAA trinucleotide repeat expansion in FXN intron 1 induces heterochromatin formation with H3K9me3 and DNA hypermethylation silencing frataxin transcription; the FXN protein coding sequence is structurally normal so a transgenic cDNA (without the expanded intron) can restore frataxin expression	source_checked
Tuberous sclerosis complex	ORPHA:805	TSC2	haploinsufficiency	conditional	gene_addition_tumour_suppressor_rescue	Germline TSC2 mutation plus somatic second-hit loss of the wild-type allele drives mTORC1 hyperactivation and hamartoma formation in multiple tissues; gene addition addresses the haploinsufficient state but cannot prevent independent somatic second-hit events in susceptible cells throughout the body	source_checked
Rett syndrome	ORPHA:778	MECP2	haploinsufficiency	conditional	aav9_cns_gene_addition_with_dosage_control	X-linked haploinsufficiency of MECP2 methyl-CpG binding protein 2; gene addition is dosage-sensitive because MECP2 duplication (three copies) causes a distinct severe neurological disease in males — therapeutic window for expression is narrow	source_checked
Dravet syndrome	ORPHA:1306	SCN1A	haploinsufficiency	conditional	lv_or_aav_inhibitory_interneuron_targeted_scn1a	Haploinsufficiency of Nav1.1 sodium channel (SCN1A) preferentially reduces firing in fast-spiking GABAergic inhibitory interneurons causing disinhibition and refractory seizures	source_checked
Fragile X syndrome	ORPHA:908	FMR1	repeat_expansion_silencing	conditional	aav_cns_fmr1_cdna_addition	CGG trinucleotide repeat expansion (>200 repeats) in the FMR1 5-prime UTR triggers de novo CpG methylation and transcriptional silencing; the FMR1 protein coding sequence is structurally normal when expressed	source_checked
Angelman syndrome	ORPHA:72	UBE3A	genomic_imprinting	conditional	aav_neuronal_ube3a_transgene_addition	Maternal UBE3A is lost by deletion or mutation; paternal UBE3A is epigenetically silenced in neurons by the SNHG14/UBE3A-ATS antisense transcript; neuronal UBE3A expression is absent from both alleles	source_checked
CDKL5 deficiency disorder	ORPHA:163934	CDKL5	haploinsufficiency	conditional	aav9_cns_cdkl5_gene_addition	X-linked haploinsufficiency of CDKL5 serine/threonine kinase disrupts neuronal cytoskeletal and synaptic signalling cascades causing early-onset refractory seizures and severe intellectual disability	source_checked
GATA2 deficiency	ORPHA:247770	GATA2	haploinsufficiency	conditional	ex_vivo_hsc_gata2_gene_addition	Autosomal dominant haploinsufficiency of GATA2 transcription factor depletes haematopoietic stem cells and lymphatic endothelial progenitors causing combined immunodeficiency bone marrow failure and lymphedema	source_checked